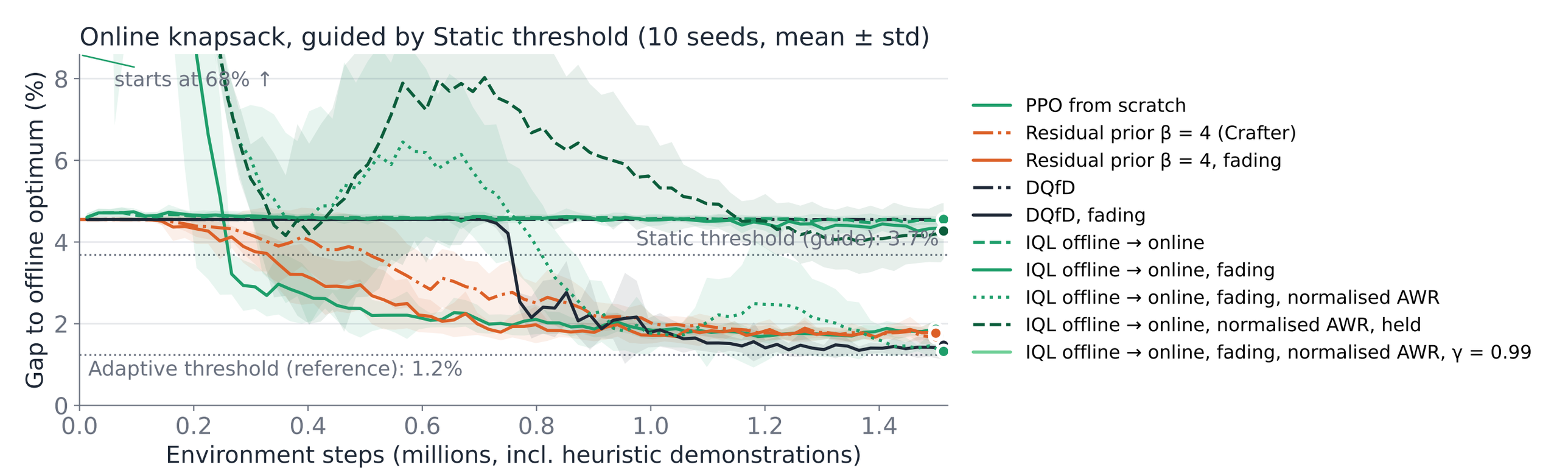
Online knapsack, guided by Static threshold (10 seeds, mean ± std)
starts at 68% ↑
Static threshold (guide): 3.7%
Adaptive threshold (reference): 1.2%
Gap to offline optimum (%)
0
2
4
6
8
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Environment steps (millions, incl. heuristic demonstrations)
PPO from scratch
Residual prior β = 4 (Crafter)
Residual prior β = 4, fading
DQfD
DQfD, fading
IQL offline → online
IQL offline → online, fading
IQL offline → online, fading, normalised AWR
IQL offline → online, normalised AWR, held
IQL offline → online, fading, normalised AWR, γ = 0.99